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# UAV Georeferencing for High-Precision Spatial Localization of Surfacing Yangtze Finless Porpoises

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## Highlights

### What are the main findings?

- LapConv, a plug-and-play Laplacian edge-enhancement module inserted into the backbone of each YOLO-OBB detector, consistently improved mean mAP@0.5 across four YOLO-OBB detector architectures, with gains ranging from 0.019 to 0.026.
- Compared with direct georeferencing, AAT-BA reduced the geometric georeferencing RMSE of static verification points from 2.59 to 0.62 m under straight-flight conditions and from 3.61 to 0.67 m under turning-flight conditions.

### What are the implications of the main findings?

- The integrated UAV framework enables high-precision spatial monitoring of small aquatic wildlife in low-texture water environments where conventional georeferencing fails, and is potentially transferable to other freshwater cetacean species.
- KDE mapping characterized differences in the spatial distribution of detected porpoises among the three survey subregions, including a left-bank concentration in Kangshan.

## Abstract

Accurate spatial localization of small, transient targets in low-texture aquatic environments remains a fundamental challenge in UAV-based remote sensing, where open-water surfaces often lack stable tie points, degrading exterior orientation estimation and conventional photogrammetric georeferencing. An integrated UAV framework combining DG/AAT-BA georeferencing with deep-learning-based oriented bounding box (OBB) detection was implemented for high-precision localization, validated on the Critically Endangered Yangtze finless porpoise (YFP, Neophocaena asiaeorientalis) in the Yangtze–Poyang Lake system. The georeferencing component selects direct georeferencing (DG) in open-water scenes and automated aerial triangulation with bundle adjustment (AAT-BA) in feature-rich nearshore scenes. Validation using two static verification points showed that, relative to DG, AAT-BA reduced geometric georeferencing RMSE from 2.59 to 0.62 m under straight-flight conditions and from 3.61 to 0.67 m under turning-flight conditions. For target detection, a lightweight Laplacian edge-enhancement convolution module (LapConv) was incorporated into YOLO-OBB backbones, amplifying weak-edge and low-contrast features of partially submerged targets. Across four representative YOLO-OBB models and three group-constrained partitions, LapConv consistently improved the mean mAP@0.5, with

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gains of 0.026, 0.024, 0.019, and 0.026 for YOLOv8, YOLO11, YOLO12, and YOLO26, respectively. Applying this framework to six UAV missions across three ecologically and hydrologically distinct subregions enabled georeferenced mapping of porpoise distributions and visualized spatial distribution characteristics during the survey period. The approach is reproducible, minimally invasive, and potentially transferable to UAV-based monitoring of other small aquatic wildlife, providing a methodological basis for fine-scale spatial surveys and subsequent habitat analysis.

**Keywords:** UAV remote sensing; oriented bounding box detection; small-target detection; Yangtze finless porpoise; wildlife monitoring

## 1. Introduction

Unmanned aerial vehicles (UAVs) equipped with high-resolution imaging systems have become an established platform for wildlife surveys, environmental monitoring, and spatial data acquisition across diverse landscape types [1–3]. In well-textured terrestrial environments, the combination of high-resolution imagery, GNSS/IMU-derived position and orientation data, and standard aerial triangulation or direct georeferencing supports high geometric accuracy and reliable spatial information extraction [4–6]. Aquatic environments, however, pose substantially greater challenges. Open-water surfaces typically lack stable and repeatable features, which weakens tie-point support, destabilizes image matching, and degrades the optimization of exterior orientation parameters in conventional photogrammetric workflows [7,8]. Although ground control points (GCPs) remain the most reliable means of achieving high absolute accuracy in UAV photogrammetric workflows, their placement is infeasible over open water surfaces; studies evaluating the effect of GCP configuration on mapping accuracy have confirmed that the absence or reduction in GCPs introduces systematic positional errors that cannot be compensated by onboard GNSS alone [9,10].These limitations become particularly acute when targets are small, sparsely distributed, and visible only transiently at the water surface—conditions that characterize the surfacing behavior of small freshwater cetaceans and many other aquatic wildlife species.

The Yangtze finless porpoise (Neophocaena asiaeorientalis, YFP) exemplifies this class of challenging targets. As the world's only freshwater porpoise, the Yangtze finless porpoise is listed as Critically Endangered by the IUCN and as a Class I nationally protected species in China [11,12]. As surfacing locations of YFPs reflect clear habitat selectivity driven by hydrodynamic conditions, prey availability, nearshore geomorphology, and anthropogenic disturbance [13–15], high-precision spatial data on surfacing events are essential for characterizing habitat use, identifying activity hotspots, evaluating responses to disturbance, and supporting population monitoring and conservation management. Its regular breathing behavior produces brief but detectable surface appearances, providing a behavioral basis for image-based detection and spatial localization [16–18]. Traditional porpoise surveys have largely relied on vessel-based visual observations, in which trained observers record sighting positions with GPS and compile individual counts [19,20]. However, vessel presence and associated underwater noise may induce avoidance behavior and alter habitat use, while boat-based surveys can have limited accessibility and detection coverage in shallow shoals, tributaries, and structurally complex nearshore waters [21,22]. By contrast, UAVs can acquire high-resolution imagery from higher altitudes, enabling rapid coverage of target waters while reducing disturbance to the animals and providing new technical support for the fine-scale acquisition of porpoise surfacing information [23,24].

Advances in UAV platforms, high-resolution imaging, and deep-learning detection have accelerated automated wildlife identification from visible-light imagery [25]. Object detection has shown substantial potential for improving monitoring efficiency, reducing manual interpretation effort, and expanding survey coverage [26–28]. Several YOLO-based frameworks have recently been developed for wildlife detection from aerial or UAV imagery. For example, Chen et al. introduced YOLO-SAG to address multi-scale variation in wildlife detection [29], while Gao et al. further improved small-target UAV wildlife surveys through thermal-RGB fusion in FDM-YOLO [30]. Despite these advances, automated detection remains particularly challenging for small freshwater cetaceans of this type living in aquatic environments, such as the Yangtze finless porpoise (YFP). In UAV imagery, surfacing individuals usually occupy only a very small proportion of the frame. Their brief and partial exposure above the water surface, together with wind-driven ripples, water turbidity, and specular reflection, often weakens target boundaries and reduces foreground-background contrast, thereby increasing the likelihood of missed detections [31,32]. In addition, differences in turbidity, algal presence, aquatic vegetation, and surface texture across habitats can generate substantial domain shifts between training and application scenes [33,34]. Despite recent advances in deep-learning-based animal detection from aerial imagery, the detection of small cetaceans in UAV images remains challenging because surfacing individuals are typically small, transient, and only partially visible [35]. Variations in water color, illumination, foam, and surface texture can further reduce target–background contrast and affect detection robustness across aquatic scenes [36].

In this study, we propose and validate an integrated UAV-based framework for fine-scale monitoring of surfacing YFPs. The contributions are threefold:

(1) A flight-segment-based hybrid georeferencing method for heterogeneous water environments. We introduce a unified spatial projection formulation that switches between direct georeferencing (DG) in open-water scenes, where tie-point matching is ill-posed, and automated aerial triangulation with bundle adjustment (AAT-BA) in feature-rich nearshore scenes, where stable land features support exterior-orientation optimization.

(2) A plug-and-play edge-enhancement module for weakly exposed small aquatic targets. We propose LapConv, a lightweight Laplacian edge-enhancement convolution module that can be integrated into the early backbone of any modern YOLO-OBB detector. Benchmarked across four representative YOLO-OBB detector architectures (YOLO8n, YOLO11n, YOLO12n, and YOLO26n) using three group-constrained holdout partitions and two training seeds per partition, LapConv consistently improved the mean mAP@0.5 across all four tested backbones.

(3) End-to-end validation in three ecologically representative habitats. We conducted six UAV missions across the Anqing mainstem, Duchang open-lake, and Kangshan river-mouth subregions of the Yangtze–Poyang Lake system, quantified localization accuracy on repeated observations of static verification points spanning straight- and turning-flight conditions, and demonstrated the applicability of the framework through georeferenced mapping and hotspot visualization of detected surfacing events across the three subregions.

The remainder of this paper is organized as follows. Section 2 describes the study area and data acquisition. Section 3 details the dataset construction, the flight- segment -based hybrid georeferencing method, the LapConv module, and its integration with YOLO-OBB detectors. Section 4 presents detection and localization results. Section 5 discusses the findings in relation to recent UAV remote sensing and conservation literature, addresses limitations, and outlines conservation implications.

## 2. Materials

### *2.1. Study Area*

The study was conducted in the Yangtze–Poyang Lake system, located in the middle and lower reaches of the Yangtze River across Anhui and Jiangxi provinces, China. This area is a primary habitat and one of the most important strongholds for the remaining population of the critically endangered Yangtze finless porpoise (Neophocaena asiaeorientalis), providing essential resources for feeding, resting, and social interaction [37]. The broader water system includes the mainstem river, transitional river–lake zones, and extensive open-lake areas, thereby offering strong contrasts in water depth, flow regime, and shoreline setting. Such environmental diversity is well suited to capturing variation in habitat preference and activity patterns. To evaluate the robustness and generalizability of the proposed method under different environmental conditions, three representative subregions were selected: Anqing, Duchang, and Kangshan (Figure 1). Together, these sites represent the major ecological and hydrological settings of the Yangtze–Poyang Lake system and provide a stratified basis for testing detection and spatial localization performance across heterogeneous water landscapes.

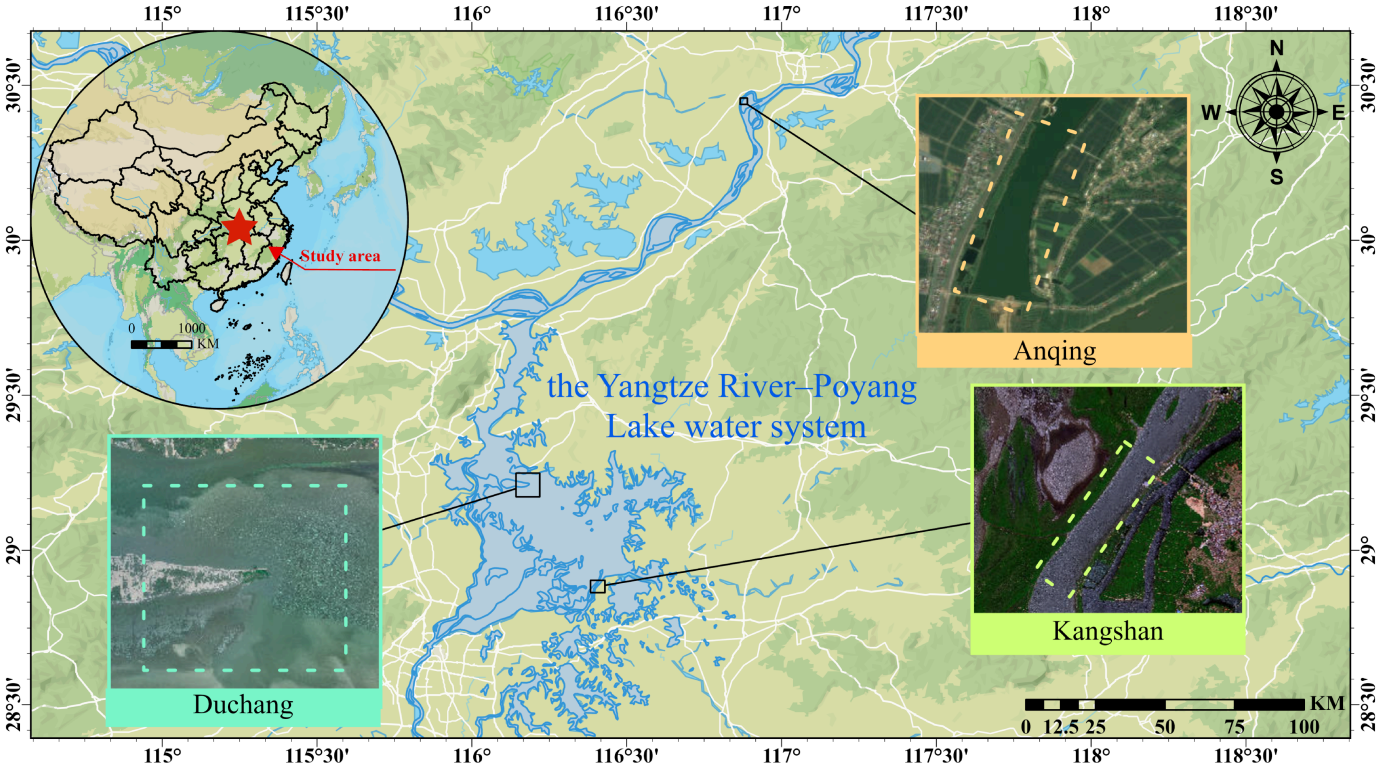


**Figure 1.** Location of the study area and the three representative survey subregions.

The Anqing subregion lies within the Anqing Xijiang Nature Reserve in Anhui Province and includes a relatively enclosed, sinuous tributary segment of the Yangtze River (30.45–30.55° N, 116.95–117.05° E). The channel has a clearly defined mainstem and a winding planform, providing a relatively stable and confined aquatic environment for porpoises. The Duchang subregion is located in the north-central part of Poyang Lake (29.30–29.45° N, 116.15–116.30° E) and is characterized by open water with weak shoreline constraints and gently curved bays and channels, producing a largely unconfined habitat. The Kangshan subregion, situated along the Kangshan River flowing into Poyang Lake from south to north (29.20–29.30° N, 116.35–116.50° E), represents a semi-enclosed riverine environment influenced by tributaries, nearshore landforms, and floodplains, with moderate channel meandering and alternating narrow and wide sections. These three

subregions differ substantially in water connectivity, shoreline geometry, channel structure, and nearshore landscape elements, and therefore provide a stratified test framework for evaluating the robustness and generalizability of the detection and spatial localization algorithms under heterogeneous water conditions.

### *2.2. Data Acquisition and Preprocessing*

UAV surveys were conducted from 11 May to 13 May 2025, using a fixed-wing CW-15 UAV equipped with a CA-103 digital aerial camera to acquire high-resolution visible-light imagery (Figure 2). To ensure consistency among missions while balancing target visibility, operational efficiency, and minimal disturbance to porpoises, flight parameters were standardized as follows: an effective flight altitude of approximately 150 m above the water surface, corresponding to a nominal ground sampling distance (GSD) of approximately 1.61 cm pixel$^{-1}$, a cruise speed of approximately 16 m/s, an image capture interval of approximately 1 s, and a forward overlap of 80–90% between consecutive images. Each image was accompanied by synchronized POS (Position and Orientation System) data to support subsequent geometric correction and precise spatial localization of surfacing porpoises. This flight altitude provided sufficient spatial resolution for porpoise detection while maximizing image coverage and minimizing potential disturbance from UAV noise. Short capture intervals and high image overlap also increased temporal sampling density, allowing more continuous recording of transient surfacing events and reducing the likelihood of missed detections.

| CA103 Specification | |
|---|---|
| Sensor size | 35.7 × 23.8 mm (Full-frame) |
| Effective pixels | 61 MP |
| Image resolution | 9504 × 6336 px |
| Pixel size | ~3.76 µm |
| Shutter speed | 1/8000 s – 30 s |

**Figure 2.** UAV platform and payload system used for high-resolution imagery acquisition. The fixed-wing CW-15 UAV is equipped with a CA-103 digital aerial camera for visible-light imaging.

Following each flight, raw UAV POS data were post-processed in JoPPS software (version 1.0). Flight-control attitude files, base-station observations, and rover measurements were jointly processed while accounting for antenna–camera lever-arm offsets, base-station coordinates, and instrument height. Corrections for exposure-time delay and antenna phase-center offset were applied to reduce systematic biases in the image-center positions and exterior orientation parameters. The processed POS data were exported in WGS84 coordinates and used as the basis for image geometric correction and porpoise spatial localization. The corrected POS data were then imported into DJI Terra, where all subsequent photogrammetric and geospatial processing steps, including image alignment,

automated aerial triangulation, bundle adjustment, orthophoto generation, and related analyses, were performed.

A total of six UAV missions were completed: three in Duchang, two in Kangshan, and one in Anqing (Table 1). Although the missions differed in acquisition time, number of images, and coverage area, the combined dataset provided effective coverage of the study sites and captured temporally continuous and environmentally diverse imagery for subsequent detection and spatial analysis of Yangtze finless porpoises.

**Table 1.** Summary of UAV survey flights and image acquisition across the study regions.

| Water System | Region | Flight ID | Number of Photos | Duration | Coverage (ha) |
|---|---|---|---|---|---|
| Poyang Lake | Duchang | D-1 | 4648 | 05:47~06:49 | 873.60 |
| | | D-2 | 4391 | 08:05~09:04 | 734.97 |
| | | D-3 | 3189 | 10:53~11:42 | 430.72 |
| | Kangshan | K-1 | 2723 | 07:05~08:02 | 300.61 |
| | | K-2 | 3661 | 10:14~11:03 | 81.1 |
| Yangtze Mainstream | Anqing | A-1 | 4706 | 16:18~17:32 | 55.05 |

## 3. Methods

### *3.1. Dataset Construction*

A total of 23,318 original high-resolution UAV images were acquired during the survey missions. All images were first manually screened based on image content and valid water-surface coverage within the surveyed areas. Images dominated by terrestrial areas or lacking valid water-surface coverage were excluded. Among the remaining images, 329 source images were identified as containing visible Yangtze finless porpoises, comprising 446 annotated porpoise instances. Porpoise-containing images accounted for only approximately 1.41% of the full UAV image collection. Moreover, even when porpoises were successfully recorded in the imagery, they occupied only a very small proportion of the original 9504 × 6336-pixel frames, as illustrated in Figure 3a. The positive source images were therefore cropped into 1024 × 1024-pixel tiles, and only tiles containing identifiable porpoise targets together with their surrounding water-surface context were retained. Some tiles contained more than one porpoise, resulting in 446 annotated instances distributed across 391 positive tiles. The remaining water-surface imagery contained a large number of highly repetitive homogeneous-background regions. Retaining all target-free tiles would not only produce a severe imbalance between positive and negative samples but also cause the dataset to be dominated by low-information backgrounds. A preliminary YOLO detector was therefore trained on the initial positive samples and applied to the remaining images with valid water-surface coverage to identify candidate hard-negative samples. The resulting false-positive detections were manually reviewed, revealing four major categories of visually confusing interference: non-target fish, sun glint, floating algae, and birds. A total of 257 representative hard-negative tiles were selected from these categories. The preliminary model was used exclusively to support hard-negative discovery and was not included in the final model evaluation. The resulting detection dataset comprised 648 tiles of 1024 × 1024 pixels, including 391 positive tiles and 257 hard-negative tiles.

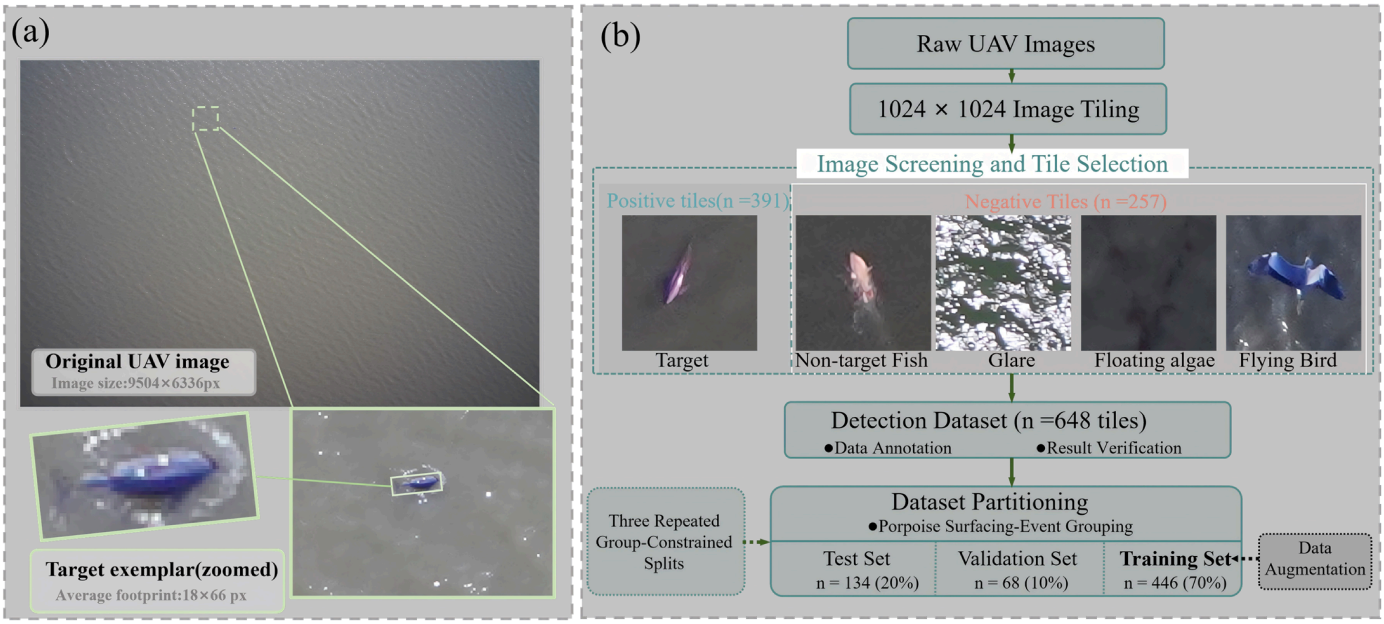


**Figure 3.** Characteristics of the UAV dataset. (**a**) An example of an original UAV image (9504 × 6336 pixels) and its corresponding 1024 × 1024 zoomed tile, showing the extremely small proportion of porpoise targets in a single frame. (**b**) Workflow for positive-tile construction, hard-negative selection, group-constrained dataset partitioning, and training-only data augmentation.

Short image-capture intervals and high forward overlap meant that temporally adjacent images could record the same porpoise surfacing event or contain highly similar water-surface backgrounds. A conventional random split at the individual-tile level could therefore distribute strongly correlated samples from the same surfacing event, source image, or continuous observation sequence across different subsets, leading to data leakage and overly optimistic performance estimates. To reduce the risk of data leakage, positive samples were grouped before dataset partitioning based on field sighting records, acquisition timestamps, and the continuity of the original image identifiers. Tiles associated with the same surfacing event or continuous observation sequence were assigned to a common group, while multiple tiles cropped from one original image were always kept together. The 391 positive tiles were ultimately organized into 121 distinct porpoise-sighting event groups. Hard-negative samples were similarly grouped according to their source images and continuous flight sequences to prevent highly similar background patterns from being distributed across different subsets. Dataset partitioning was subsequently performed at the event-group and source-sequence levels, ensuring that no surfacing event, original image, or continuous image sequence occurred in more than one subset.

To reduce the sensitivity of model evaluation to a single split under limited-data conditions, three repeated group-constrained holdout partitions were generated while preserving the integrity of all event and source-sequence groups. Each partition followed an approximate 7:1:2 ratio and contained 446 training tiles, 68 validation tiles, and 134 test tiles. The same three partitions were used for all evaluated models to ensure fair and paired comparisons. For each partition, every backbone–variant configuration was trained using two independent training seeds to account for variability arising from model initialization, data shuffling, and stochastic augmentation. Thus, each configuration was evaluated through six training runs in total. Detection metrics were first averaged across the two training seeds within each partition. The resulting partition-level metrics were then summarized as the mean and standard deviation across the three independent partitions, thereby characterizing performance variability associated primarily with differences in

subset composition. The limited number of porpoise-positive samples increased the risk of overfitting when training only on the original images and restricted the variation in target appearance available to the model [38,39]. Data augmentation was therefore applied exclusively to the training subsets after dataset partitioning. The augmentation operations included Mosaic, MixUp, rotation, scaling, translation, color perturbation, and horizontal flipping, exposing the model to greater variation in target scale, orientation, spatial configuration, illumination, and water-surface background. The validation and test subsets were retained in their original form without image duplication or augmentation, preventing transformed samples from entering the independent evaluation sets and preserving the objectivity of the reported performance estimates.

### *3.2. Flight-Segment-Based Hybrid Spatial Georeferencing Method*

UAV imagery in the study area often contains extensive low-texture water surfaces, which reduces the stability of conventional geometric orientation. To address this problem, two complementary spatial localization strategies were adopted. The method switches between two orientation strategies according to scene feature richness and then estimates three-dimensional target coordinates through a unified spatial projection model.

For open-water scenes, repeated surface textures and the lack of stable features can make feature matching ill-posed. Direct georeferencing (DG) was therefore used. DG relies on post-processed POS data to obtain the exterior orientation elements directly:

$$\mathbf{E}_i^{\mathrm{DG}} = \left[X_{s,i}^{\mathrm{POS}}, Y_{s,i}^{\mathrm{POS}}, Z_{s,i}^{\mathrm{POS}}, \psi_i^{\mathrm{POS}}, \theta_i^{\mathrm{POS}}, \phi_i^{\mathrm{POS}}\right]^{\mathrm{T}} \tag{1}$$

In mixed nearshore scenes containing stable land features such as shorelines, buildings, and vegetation, high-quality tie points can be extracted. Exterior orientation elements were optimized using automated aerial triangulation (AAT) combined with bundle adjustment (BA). Let $P_j = [X_j, Y_j, Z_j]^{\mathrm{T}}$ represent the ground coordinates of the $j$-th tie point, and $(x_{ij}, y_{ij})$ its corresponding image coordinates in the $i$-th image, then the collinearity equations are satisfied:

$$x_{ij} - x_0 = -f\frac{r_{11}^{(i)}(X_j - X_{s,i}) + r_{12}^{(i)}(Y_j - Y_{s,i}) + r_{13}^{(i)}(Z_j - Z_{s,i})}{r_{31}^{(i)}(X_j - X_{s,i}) + r_{32}^{(i)}(Y_j - Y_{s,i}) + r_{33}^{(i)}(Z_j - Z_{s,i})} \tag{2}$$

$$y_{ij} - y_0 = -f\frac{r_{21}^{(i)}(X_j - X_{s,i}) + r_{22}^{(i)}(Y_j - Y_{s,i}) + r_{23}^{(i)}(Z_j - Z_{s,i})}{r_{31}^{(i)}(X_j - X_{s,i}) + r_{32}^{(i)}(Y_j - Y_{s,i}) + r_{33}^{(i)}(Z_j - Z_{s,i})} \tag{3}$$

Using POS-derived exterior orientation as the prior, the bundle-adjustment objective function can be written as:

$$\min_{\{E_i\},\{P_j\}}\left[\sum_i\sum_j\left\|p_{ij}^{\mathrm{obs}} - p_{ij}^{\mathrm{pred}}(E_i, P_j)\right\|^2 + \lambda\sum_i\left\|E_i - E_i^{\mathrm{POS}}\right\|_{W_{\mathrm{POS,j}}}^2\right] \tag{4}$$

where $p_{obs}^{ij}$ and $p_{pred}^{ij}$ are the observed and predicted image coordinates of tie points, $\boldsymbol{E}_{\mathrm{POS},i}$ is the initial exterior orientation from POS, $W_{\mathrm{POS}}$ is the weighting matrix, and $\lambda$ a balancing coefficient. The optimized exterior orientation elements are obtained as:

$$\mathbf{E}_i^{\mathrm{AT}} = \left[X_{s,i}^{\mathrm{AT}}, Y_{s,i}^{\mathrm{AT}}, Z_{s,i}^{\mathrm{AT}}, \Psi_i^{\mathrm{AT}}, \theta_i^{\mathrm{AT}}, \phi_i^{\mathrm{AT}}\right]^{\mathrm{T}} \tag{5}$$

To unify the description of different scene types, a scene indicator variable $s_i$ was introduced to select the appropriate source of exterior orientation parameters for each image. The selected exterior orientation vector $\mathbf{E}_i^*$ comprises the camera-center position

$\mathbf{S}_i^*$ and the orientation angles $\boldsymbol{\Theta}_i^*$, while the corresponding rotation matrix is calculated as $\mathbf{R}_i^* = \mathbf{R}(\boldsymbol{\Theta}_i^*)$:

$$(\mathbf{S}_i^*, \mathbf{R}_i^*) = \begin{cases} (\mathbf{S}_{\mathrm{POS},i}, \mathbf{R}_{\mathrm{DG},i}), & s_i = 0 \text{ (DG flight segment)} \\ (\mathbf{S}_{\mathrm{AT},i}, \mathbf{R}_{\mathrm{AT},i}), & s_i = 1 \text{ (AAT-BA flight segment)} \end{cases} \tag{6}$$

Here, $s_i = 0$ denotes flight segments processed using direct georeferencing (DG), whereas $s_i = 1$ represents flight segments whose exterior orientation elements were further refined using automated aerial triangulation and bundle adjustment (AAT-BA). Georeferencing mode was assigned at the flight-segment level according to land-overlap and feature-matching criteria. The land-overlap ratio was calculated as the proportion of the flight-segment image footprint intersecting adjacent land areas. Flight segments with a land-overlap ratio of at least 10% were considered eligible for AAT-BA. Feature-matching quality was then evaluated between adjacent overlapping images, and AAT-BA was applied only when at least 80 geometrically verified inlier tie points remained after RANSAC filtering. Flight segments that did not satisfy both criteria were processed using DG. Under this rule, the Duchang flight lines, which predominantly covered open water and lacked sufficient stable land features, were processed using DG. In Kangshan and Anqing, flight segments overlapping shorelines or adjacent land areas were further optimized using AAT-BA when both quantitative criteria were satisfied. For each image, $s_i$ was assigned according to the georeferencing mode of its corresponding flight segment. This unified formulation enables the 3D object-space coordinates $T_{g,ij}$ to be consistently calculated using the same spatial projection model. Finally, the 3D object-space coordinates of porpoise targets are expressed as:

$$\mathbf{T}_{g,ij} = \mathbf{S}_i^* + \frac{Z_w - Z_{s,i}^*}{\left(\mathbf{R}_i^* \mathbf{r}_{ij}^{\mathrm{cam}}\right)_z} \mathbf{R}_i^* \mathbf{r}_{ij}^{\mathrm{cam}} \tag{7}$$

where $\boldsymbol{T}_{g,ij}$ denotes the three-dimensional ground coordinates of the $j$-th detected target in image $i$, $\mathbf{S}_i^*$ is the selected camera-center position, $Z_{s,i}^*$ is the corresponding camera-center elevation, $Z_w$ is the water-surface elevation, and $\left(\mathbf{R}_i^* \mathbf{r}_{ij}^{\mathrm{cam}}\right)_z$ denotes the vertical component of the target line-of-sight vector in the object-space coordinate system. Within each flight segment, the water surface was approximated as a locally horizontal plane at elevation $Z_w$, and the target position was determined from the intersection of the camera line of sight with this plane.

### *3.3. YOLO26-Lap: Framework and Architecture*

#### 3.3.1. YOLO26 Baseline

YOLO26 was adopted as the baseline framework for surfacing-porpoise detection in UAV imagery. Relative to YOLO11, YOLO26 removes Distributional Focal Loss (DFL) from bounding-box regression, uses an end-to-end inference paradigm without non-maximum suppression (NMS), and combines ProgLoss with STAL loss to improve small-object detection accuracy, inference efficiency, and deployment compatibility [40,41]. These properties make the model well suited to UAV-based porpoise detection, where targets are small, sparsely distributed, and frequently embedded in complex water backgrounds.

As illustrated in Figure 4a, the YOLO26 architecture comprises a backbone, a neck, and a head. The backbone extracts hierarchical features through stacked convolutional layers, C3K2 modules, and SPPF blocks, generating multi-scale representations of porpoise targets and surrounding water patterns. The neck then aggregates high-resolution texture information and low-resolution semantic information through upsampling and concatenation. Finally, the head produces detection results at multiple scales, enabling simultaneous identification of porpoises with different sizes and exposure states.

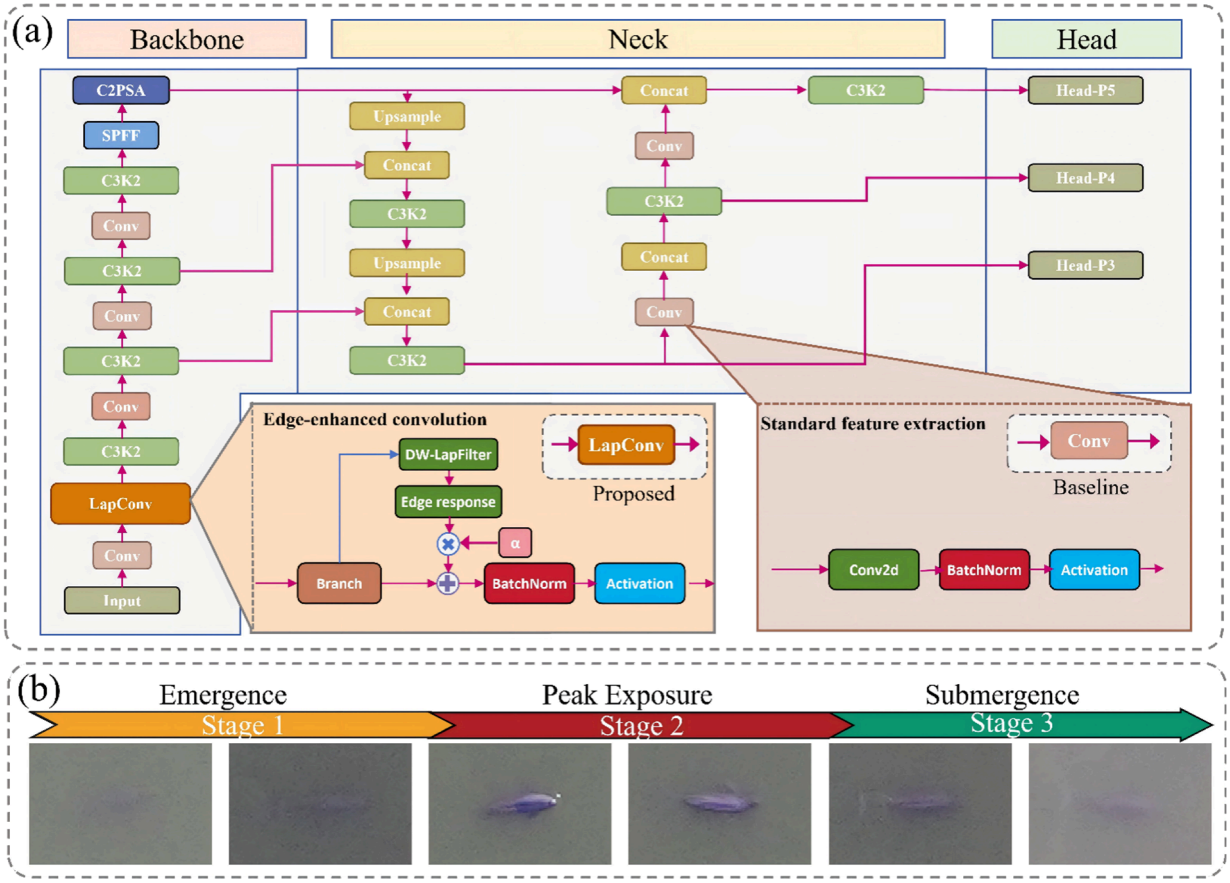


**Figure 4.** The proposed YOLO26-Lap framework and typical stages of porpoise surfacing. (**a**) Network architecture with the embedded LapConv module. (**b**) Representative samples of emergence, peak exposure, and submergence stages.

#### 3.3.2. Laplacian Edge Enhancement Convolution (LapConv)

Surfacing porpoises in UAV imagery can be divided into three typical stages (Figure 4b): emergence, peak exposure, and submergence. During peak exposure, a relatively complete portion of the body is visible, and the target boundary is distinguishable from the background. During emergence and submergence, however, only a small part of the body is exposed, and the target often exhibits weak edges, blurred contours, and low foreground–background contrast due to water turbidity, surface reflection, refraction, and light scattering. These conditions substantially increase the risk of missed detections and inaccurate bounding-box regression, particularly for oriented bounding box (OBB) detection, which relies on precise edge delineation to recover both the target's location and its longitudinal body axis.

To address this problem, a Laplacian Edge Enhancement Convolution (LapConv) module was inserted into the second convolutional layer of the backbone, as highlighted in Figure 4a. This location was selected because early layers primarily capture low-level visual cues such as edges, textures, and corners, which are especially important for small-target perception [42]. Strengthening these cues early in the network helps preserve subtle porpoise boundaries before deeper feature extraction attenuates them. In the proposed design, the input feature map is first processed with a depthwise Laplacian filter to generate an edge-response map. The enhanced edge information is then fused back into the main feature stream and followed by batch normalization and activation. This design strengthens local gradient responses while suppressing irrelevant noise.

The discrete Laplacian kernel is defined as:

$$\mathbf{L} = \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix} \tag{8}$$

Accordingly, the enhanced feature can be expressed as:

$$g(x, y) = f(x, y) + \alpha \cdot \nabla^2 f(x, y) \tag{9}$$

In implementation, for an input feature map $X$, the edge-response feature $E$ is first computed by applying a depthwise convolution with the Laplacian kernel $K_L$:

$$\mathbf{E} = \text{DepthwiseConv2D}(\mathbf{X}, \mathbf{K}_L, \text{padding} = 1) \tag{10}$$

The resulting edge-response feature is then adaptively fused with the original feature map through a learnable scalar parameter $\alpha$:

$$\mathbf{X}' = \mathbf{X} + \alpha \cdot \mathbf{E} \tag{11}$$

Here, $\alpha$ controls the edge-enhancement strength. It was initialized to 1.0 and jointly optimized with the other trainable network parameters using SGD, without an explicit range constraint or a separate learning-rate strategy. Finally, the enhanced feature is passed through a standard convolutional transformation followed by batch normalization and activation to obtain the output feature **Y**:

$$\mathbf{Y} = \text{Act}\big(\text{BN}\big(\text{Conv}\big(\mathbf{X}'\big)\big)\big) \tag{12}$$

This design allows the network to amplify weak edge and texture signals from partially submerged porpoises while remaining compatible with the original YOLO26 detection pipeline. As a result, the proposed YOLO26-Lap framework improves sensitivity to blurred and low-contrast targets, thereby improving overall detection robustness under heterogeneous water-surface conditions.

### *3.4. Workflow*

The overall workflow consists of three stages: data acquisition and preprocessing, automated detection-model construction, and spatial localization followed by result analysis (Figure 5). UAV surveys were conducted under clear and low-wind conditions, with wind speeds maintained at 3–5 m/s to ensure stable flight and to reduce water-surface ripples that could interfere with porpoise-edge detection. Each flight was typically limited to approximately 1 h, primarily by UAV endurance. To ensure flight safety and complete data acquisition, each mission was completed within the effective endurance period along the planned route. Because takeoff-site constraints inevitably caused the outbound and return segments to image areas outside the target waterbody, those images were subsequently filtered using POS data and study-area water-boundary information. Dataset construction and subsequent analyses were therefore based exclusively on imagery covering valid water surfaces.

In the spatial localization stage, processing was organized around three levels: the UAV platform, the remote-sensing image plane, and the real-world geographic plane. The core objective was to establish correspondence among UAV observation data, image-pixel positions, and real-world geographic coordinates. At the image-plane level, model training and inference were performed on the constructed dataset to detect porpoise targets in UAV imagery, and the pixel coordinates of the bounding-box center were extracted to represent target positions in the image plane. After target detection, mapping from the image plane to the real-world geographic plane was achieved by combining POS data recorded by the UAV platform with image-formation geometry. In this projection, the water

surface was approximated as a locally horizontal plane with a representative water-surface elevation. The image-space position of each porpoise was represented by the center of the oriented bounding box predicted by the deep-learning detector. Specifically, the geographic coordinates of the image center were used as the reference, and the pixel offset of the target center relative to the image center was converted into real-world ground distance using UAV flight height above the water surface together with heading and attitude parameters. The resulting offset was then projected onto the geographic plane to derive the location of each detected porpoise target. After porpoise detection and spatial localization had been completed, the framework was evaluated from three perspectives.

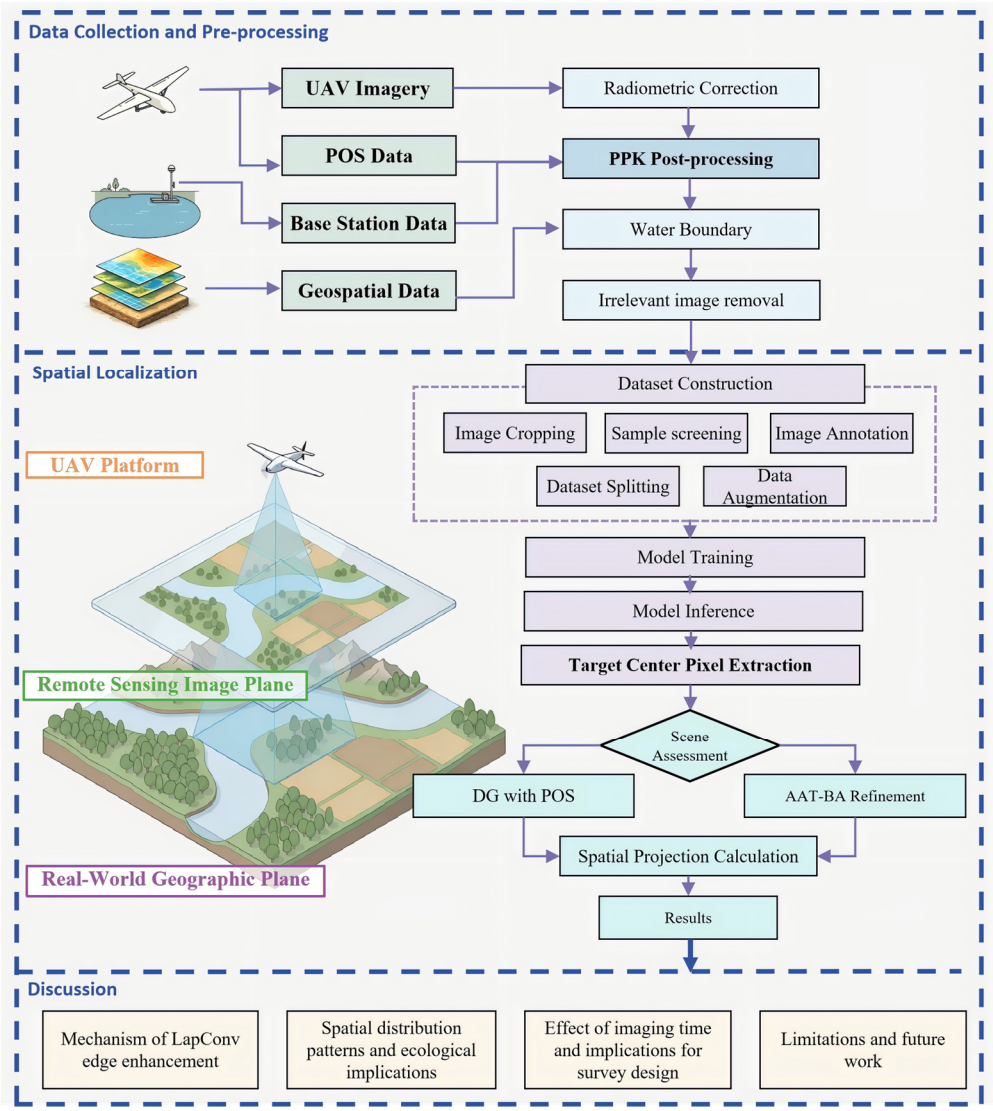


**Figure 5.** Overall workflow for UAV-based detection and spatial localization of surfacing Yangtze finless porpoises.

## 4. Results

### *4.1. Cross-Model Detection Performance*

To evaluate whether LapConv provides a consistent, backbone-agnostic improvement, we benchmarked it across four representative YOLO-OBB detectors spanning the major iterations from 2023 to 2026: YOLO8n, YOLO11n, YOLO12n, and YOLO26n. For each of the three group-constrained dataset partitions, every backbone–variant configuration was evaluated using two independent training seeds to account for training variability. All

models were implemented in PyTorch 2.5.1 using the Ultralytics framework (v8.4.2) and trained on an NVIDIA RTX 2080 Ti GPU (CUDA 11.8).

As shown in Table 2, YOLO26-Lap was further evaluated before and after data augmentation to assess the effect of the augmentation strategy on detection performance. Data augmentation increased the Precision, Recall, F1-score, and mAP@0.5 of YOLO26-Lap by 0.169, 0.097, 0.133, and 0.189, respectively. The variability of Precision, Recall, and F1-score was also reduced, although the standard deviation of mAP@0.5 increased slightly. The performance of LapConv was evaluated across four representative YOLO-OBB backbones—YOLO8n, YOLO11n, YOLO12n, and YOLO26n—on the constructed UAV dataset. As shown in Figure 6, LapConv consistently improved the mean mAP@0.5 across all four backbones. The mean mAP@0.5 increased from 0.890 to 0.916 for YOLOv8, from 0.903 to 0.927 for YOLO11, from 0.902 to 0.921 for YOLO12, and from 0.854 to 0.880 for YOLO26. The corresponding improvements were 0.026, 0.024, 0.019, and 0.026, respectively.

**Table 2.** Detection performance of YOLO26-Lap before and after data augmentation. Values are reported as mean ± standard deviation.

| Data Augmentation | Precision | Recall | F1-Score | mAP@0.5 |
|---|---|---|---|---|
| Before augmentation | 0.678 ± 0.082 | 0.688 ± 0.127 | 0.682 ± 0.102 | 0.691 ± 0.121 |
| After augmentation | 0.847 ± 0.003 | 0.785 ± 0.014 | 0.815 ± 0.009 | 0.880 ± 0.150 |

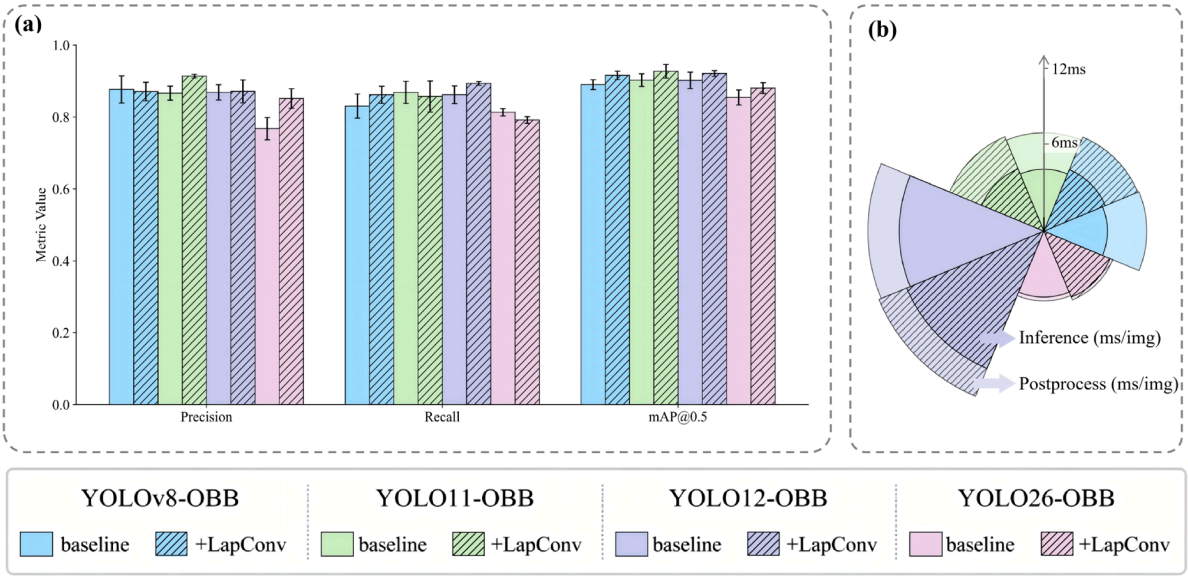


**Figure 6.** Cross-backbone comparison of detection performance and inference efficiency for baseline and LapConv-enhanced YOLO-OBB models. (**a**) Comparison of mean Precision, Recall, and mAP@0.5 across the three group-constrained partitions for four representative backbones with and without LapConv. Error bars represent the standard deviation across the three partitions. (**b**) Corresponding inference-time comparison of different backbone–variant configurations.

YOLO26 showed the largest improvement in Precision, increasing from 0.768 to 0.851, corresponding to a gain of 0.083, while its mean mAP@0.5 increased from 0.854 to 0.880. By strengthening weak edge responses and local texture representation, LapConv improves the separability of partially exposed porpoises from complex water-surface backgrounds, reduces false positives caused by ripples or floating algae, and increases the reliability of OBB detection for low-contrast, weakly exposed targets. Figure 6b illustrates the corresponding inference and post-processing times. The addition of LapConv introduces minimal computational overhead, reflecting its lightweight design. Average inference time

per image remains largely unchanged across all backbones, highlighting that the module can be applied without compromising real-time performance. Notably, YOLO26-Lap exhibits an extremely short post-processing time (0.233 ms/image), which is primarily due to its fully end-to-end, NMS-free prediction pipeline. This design eliminates the need for time-consuming post-processing steps typically required in other backbones, further enhancing the model's efficiency for real-time UAV monitoring.

As further illustrated in Figure 7, YOLO26-Lap successfully identified porpoises under different levels of edge clarity and various water-surface conditions. Although YOLO11-Lap achieved the highest mean mAP@0.5, the compact architecture, substantial Precision improvement, and short post-processing time of YOLO26-Lap make it suitable for potential real-time and resource-constrained applications. Representative detection examples illustrate successful identification of porpoises under different levels of edge clarity and different water-surface conditions. In relatively clear target scenes, the predicted bounding boxes closely matched target contours, as shown in Observation 1 and Observation 4. In Observation 3, the model also detected a local porpoise target despite background interference from surface glare, wave-induced textures, and filamentous patterns. The enlarged views further show strong agreement between the predicted boxes and actual porpoise positions, indicating that YOLO26-Lap adapts well to small-object detection under complex aquatic backgrounds.

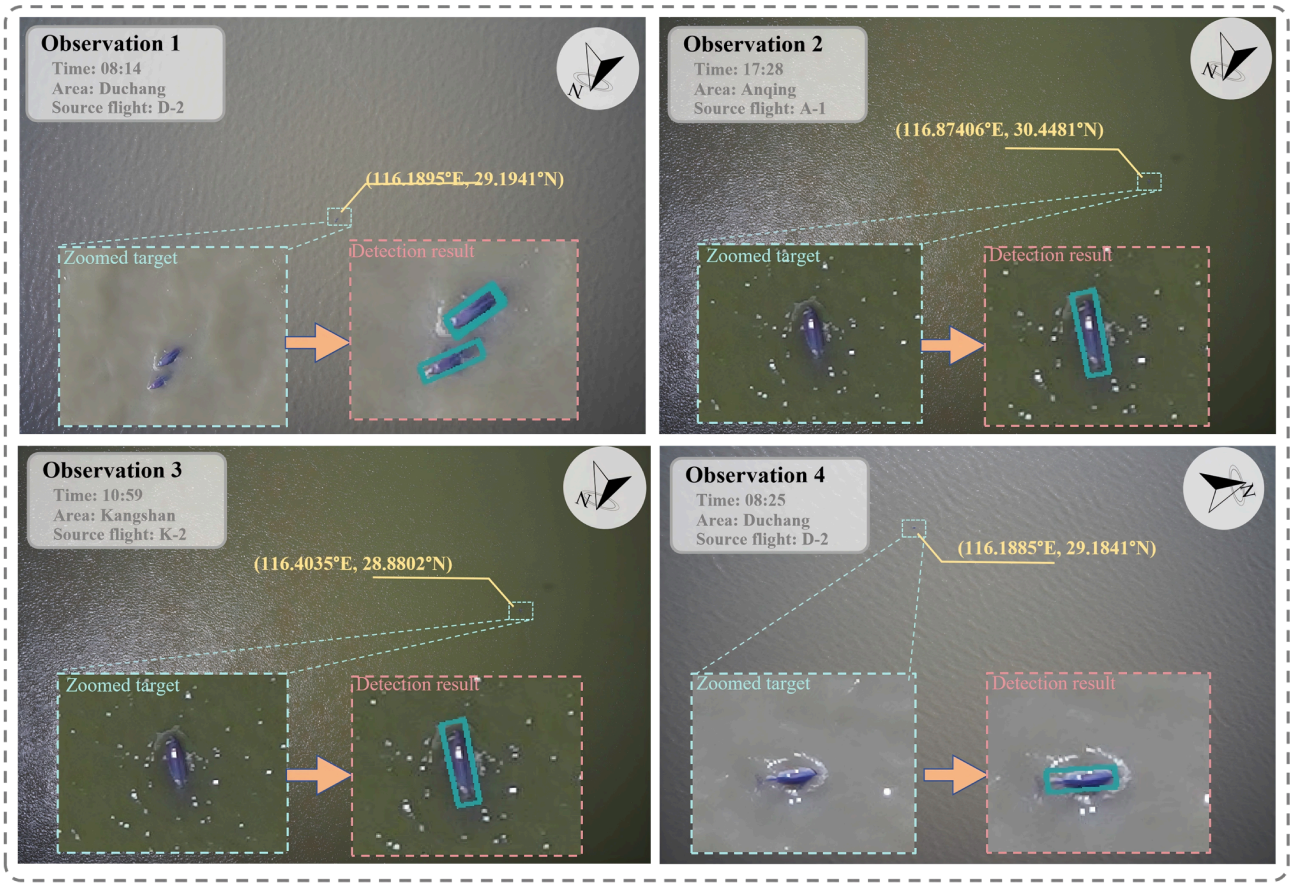


**Figure 7.** Representative examples of Yangtze finless porpoise detection and spatial localization using YOLO26-Lap in UAV ortho-rectified image fragments.

### *4.2. Localization Accuracy*

Direct use of detected porpoise positions for accuracy assessment would confound true animal movement with processing-induced error because surfacing events are highly transient and individuals may continue moving during exposure. To evaluate the positional

accuracy of the proposed localization workflow under more controlled conditions, two static verification points in the Anqing subregion were selected. Each verification point was repeatedly observed in seven UAV images, yielding seven localization samples for VP1 and seven for VP2. These reference points had fixed positions, clear visual features, and repeated visibility across overlapping images, providing a basis for quantifying the dispersion and radial errors of the geometric localization results under different flight conditions.

As shown in Figure 8a, the two reference points were deliberately selected from different flight conditions. Verification Point 1 (VP1) was observed mainly during straight flight, when UAV attitude remained relatively stable and image overlap was more consistent. VP1 therefore served as a baseline case under comparatively favorable imaging geometry. Verification Point 2 (VP2), by contrast, was observed mainly during turning flight, when stronger heading and attitude variation introduced more complex imaging geometry and greater georeferencing uncertainty. The scatter plots in Figure 8b illustrated these differences. Before AAT-BA, reconstructed positions for both points were more dispersed, particularly for VP2, reflecting the greater positional instability associated with turning flight. After AAT-BA, the point clouds for both verification points became much more concentrated, indicating that the optimized orientation parameters effectively constrained positional deviation under both straight-flight and turning-flight conditions.

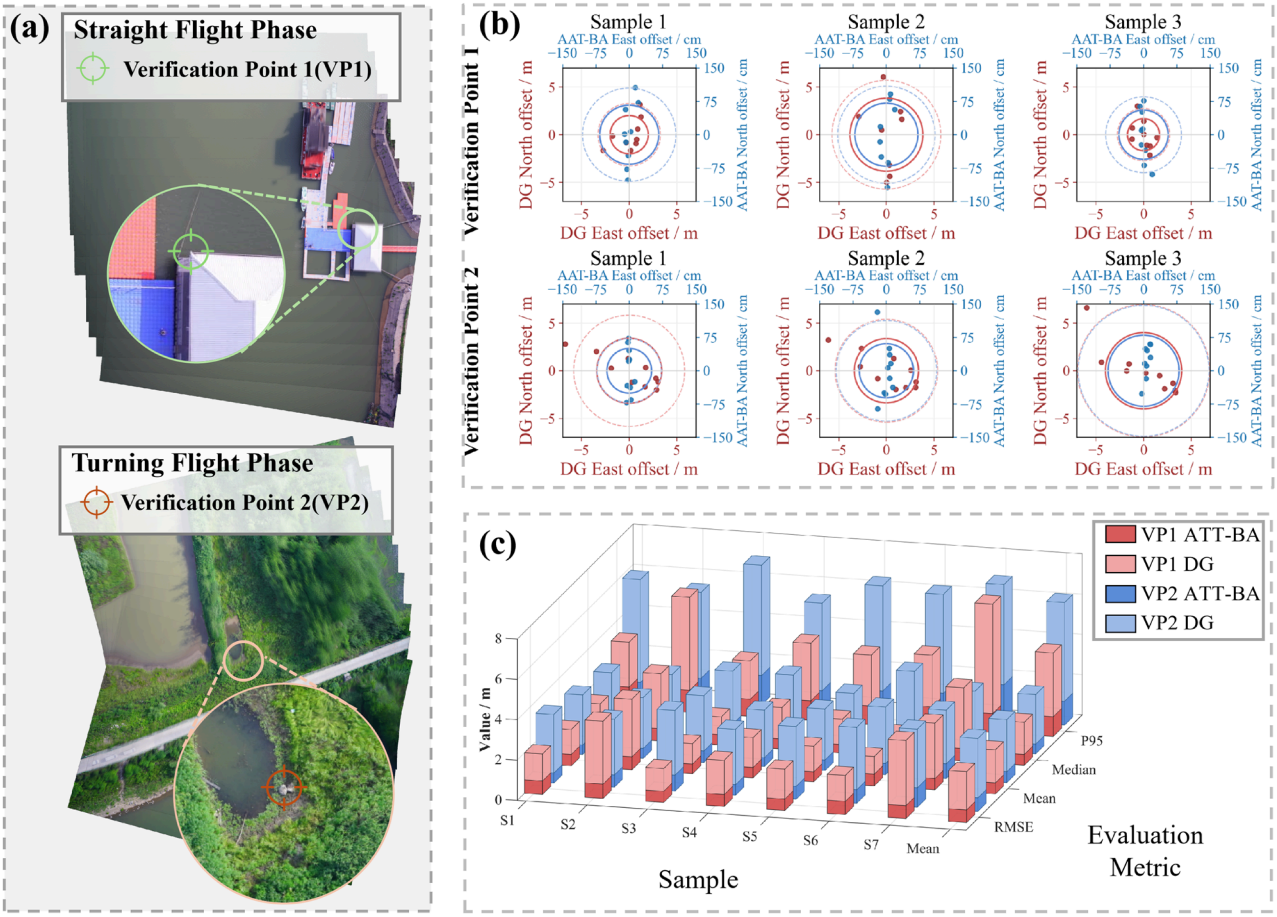


**Figure 8.** Accuracy assessment of spatial localization under different flight conditions. (**a**) Locations of two static verification points selected from straight-flight and turning-flight phases. (**b**) Offset distributions of repeated observations before and after AAT-BA for the two verification points. (**c**) Comparison of localization error metrics, including RMSE, mean, median, and P95, for different samples and verification points.

The statistical results in Figure 8c confirm the improvement quantitatively. For VP1, RMSE, mean, median, P95, and maximum radial error were 2.59 m, 2.26 m, 2.22 m, 4.21 m, and 4.84 m before AAT-BA and were reduced to 0.62 m, 0.54 m, 0.53 m, 0.97 m, and 1.01 m after AAT-BA, respectively, indicating a clear shift from meter-level to sub-meter accuracy. VP2, which represents turning-flight conditions, showed larger pre-AAT-BA errors, with RMSE, mean, median, P95, and maximum radial error of 3.61 m, 3.12 m, 2.94 m, 6.09 m, and 7.62 m, respectively. After AAT-BA, these values decreased markedly to 0.67 m, 0.54 m, 0.49 m, 1.19 m, and 1.51 m. Although post-AAT-BA errors for VP2 remained slightly larger than those for VP1, the overall reduction still demonstrates the effectiveness of AAT-BA under more challenging imaging geometry.

Given the distinct flight geometries represented by VP1 and VP2, their error statistics were analyzed separately. Relative to DG, AAT-BA reduced RMSE by 76.1% at VP1 under straight-flight conditions and by 81.4% at VP2 under turning-flight conditions. Although the post-AAT-BA RMSE values were comparable, VP2 exhibited higher P95 and maximum radial errors, indicating greater upper-tail uncertainty during turning flight. Overall, AAT-BA improved the geometric stability and consistency of repeated static-reference-point localization under both flight conditions. These results quantify the geometric component of spatial-localization uncertainty and provide a practical reference for evaluating the expected localization error of surfacing porpoises in UAV imagery.

## 5. Discussion

### *5.1. Mechanism of LapConv Edge Enhancement*

The placement of LapConv within the YOLO26 backbone was investigated through a layer-insertion ablation study, in which the module was inserted at different backbone stages (layer0, layer1, layer3, and layer5). Figure 9 summarizes the performance differences in these variants relative to the baseline YOLO26-OBB.

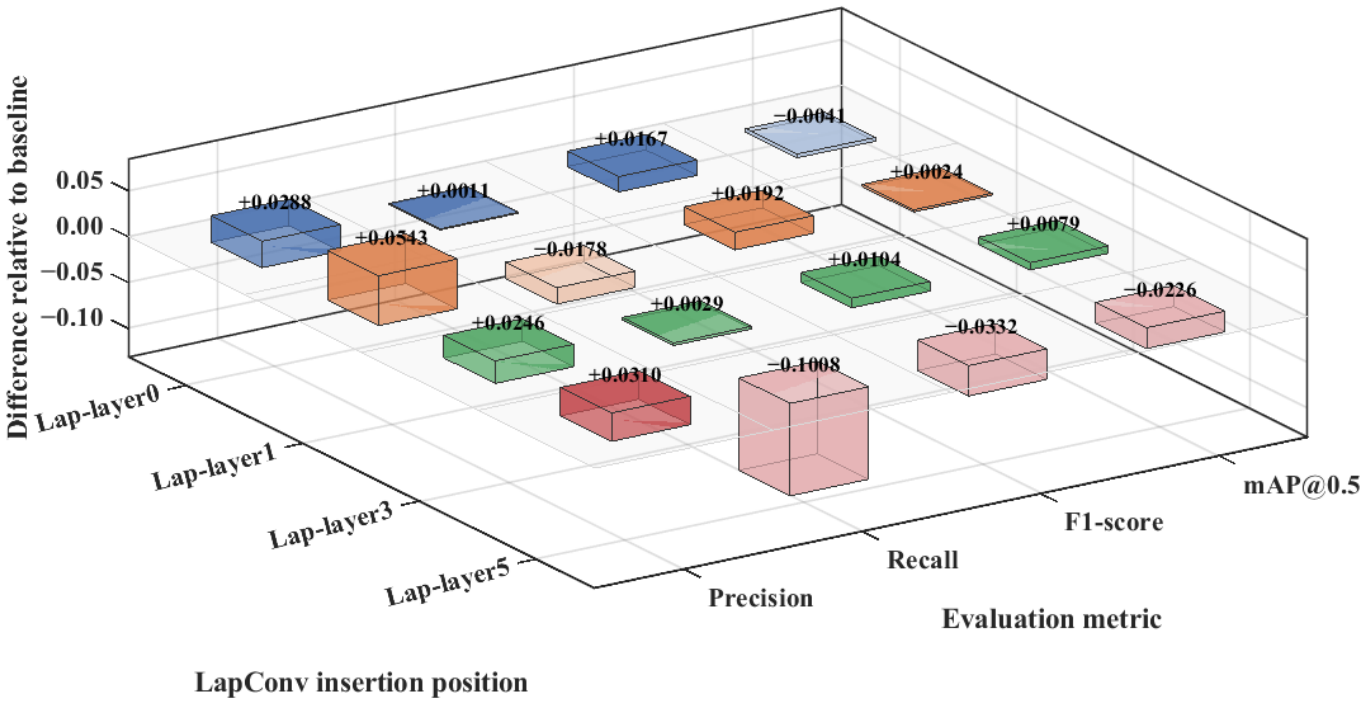


**Figure 9.** 3D bar chart of the performance differences in Precision, Recall, F1-score, and mAP@0.5 for different LapConv insertion positions relative to the YOLO26-OBB baseline.

The standard YOLO26-Lap configuration, with LapConv inserted at layer1, achieved the best overall balance of performance among the tested variants. The layer0 variant achieved the highest Recall (0.902), but its overall detection quality—including Precision, F1-score, and mAP@0.5—was lower than that of layer1, indicating that maximizing Recall alone does not produce the most balanced result. Deeper insertion variants, particularly layer5, offered smaller improvements or even reduced performance relative to the baseline. These results suggest that inserting LapConv into shallow backbone layers preserves low-level edge and texture information while enhancing detection of partially exposed targets. Overall, considering all key metrics, layer1 provides the most balanced and robust configuration for UAV-based porpoise detection, whereas layer0 may be preferred only if Recall is the sole priority.

The visual comparisons in Figure 10 help explain these quantitative differences. In the missed-detection example, two porpoises are present in the image, corresponding to the peak-exposure and submergence stages, respectively. The peak-exposure individual has a clearer contour and stronger contrast with the surrounding water, whereas the submerging individual is mostly below the water surface and therefore appears much blurrier. Under this condition, the baseline model responds weakly to the submerging target and fails to detect it. By contrast, YOLO26-Lap produces a stronger and more concentrated response in the blurred target region, enabling successful detection of both individuals. In the correct-identification example, both models detect the porpoise targets, and their feature responses are largely similar, indicating that LapConv does not significantly alter representations when the target boundaries are clear. In the false-detection example, the confusing object is not a porpoise but a small group of clustered fish. Because individual fish are small and closely aggregated, the baseline model tends to merge their boundaries and misinterpret the mixed outline as a porpoise target, leading to a false alarm. The improved model shows a more discriminative response pattern and better separates individual fish signatures within the cluster, thereby avoiding this false detection. These examples show that Laplacian edge enhancement primarily strengthens weak and blurred porpoise boundaries and improves discrimination between true porpoise targets and visually similar interference in complex water scenes, while having minimal impact on well-exposed targets.

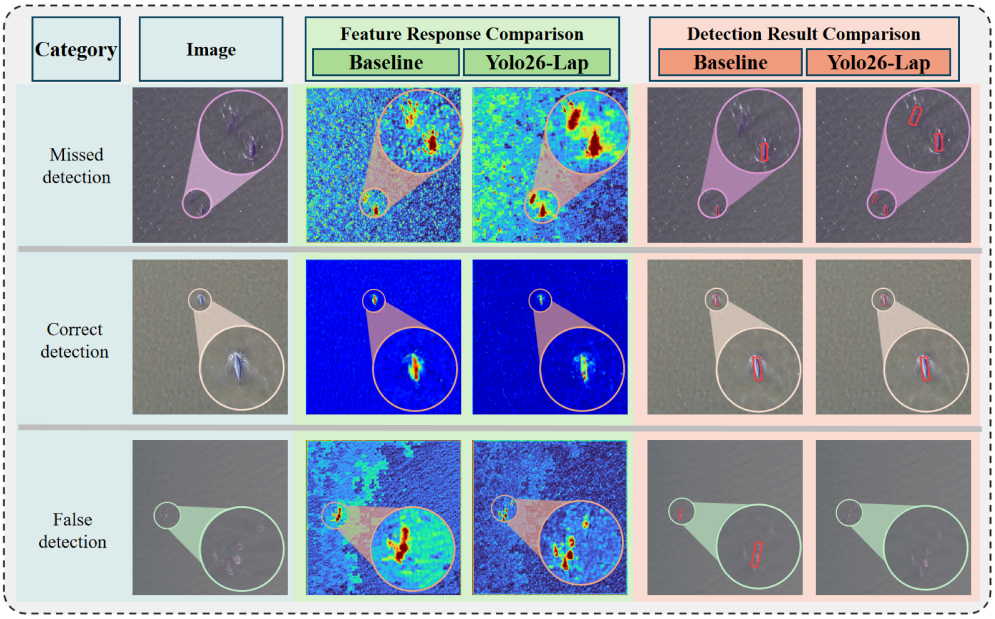


**Figure 10.** Comparison of feature responses and detection outputs between the baseline model and the proposed model under missed detection, correct identification, and false detection cases.

### *5.2. Spatial Distribution Characteristics Across Survey Subregions*

The proposed detection and spatial localization workflow was applied to all UAV flight missions across the three study subregions—Duchang, Kangshan, and Anqing. Detection results were first summarized by mission, revealing differences in mission-level porpoise detection patterns among the three subregions (Figure 11). Overall, the Anqing Nature Reserve exhibited the highest probability of detection, with mission A-1 achieving the maximum porpoise-image ratio (2.87%) and the highest porpoise density (1.57 porpoises per porpoise-containing image). By contrast, Duchang and Kangshan showed lower mean porpoise-image ratios (0.44% and 0.63%, respectively) and densities (1.17 and 1.05), reflecting lower porpoise-image ratios and more dispersed detection patterns during the surveyed missions in the natural water environments. These results suggest that enclosed and protected habitats, such as Anqing, support higher aggregation intensity and more frequent surfacing activity compared to open-water or riverine subregions.

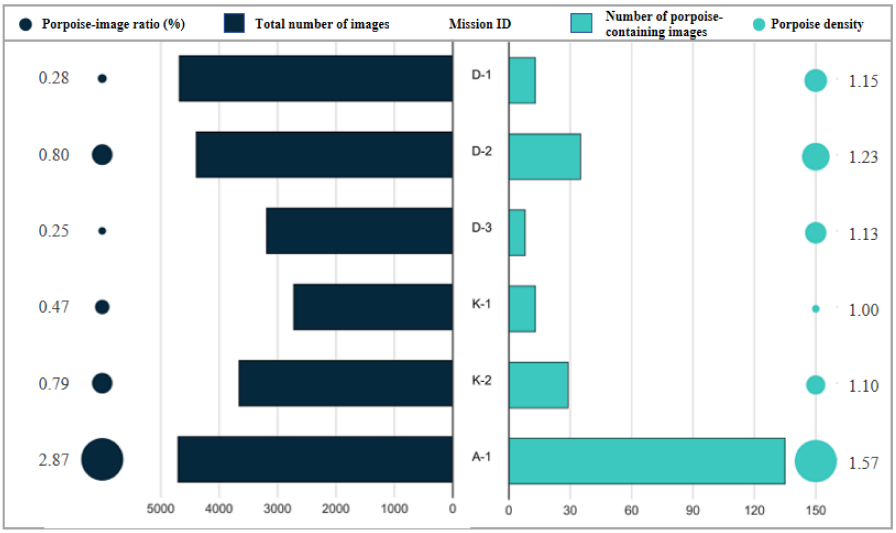


**Figure 11.** Summary statistics of porpoise detection results. The porpoise-image ratio represents the percentage of images containing at least one detected porpoise, while porpoise density denotes the average number of porpoises per porpoise-containing image.

Geographic locations of surfacing porpoises were reconstructed by combining UAV-based detections with spatial projection of target-center positions (Figure 12). The three subregions displayed different spatial distribution characteristics during the surveyed missions. In Duchang, detections were relatively dispersed across the eastern and central parts of the open-water area, with several localized clusters. In Kangshan, detections were distributed mainly along the river channel and occurred more frequently near the left bank, forming a relatively linear pattern. In Anqing, detections were more continuously distributed along the enclosed river segment and formed an elongated concentration belt. The KDE maps further visualized these spatial differences. Duchang showed several localized areas of higher detection concentration along the curved open-water corridor. Kangshan exhibited a series of longitudinal concentration areas near the left bank, whereas Anqing showed a relatively continuous high-concentration belt along the central channel. These results demonstrate that the proposed workflow can transform UAV detections into geographically explicit spatial information and describe fine-scale differences in the distribution of surfacing porpoises across contrasting aquatic environments. The georeferenced porpoise distribution records generated by this workflow provide a data basis for subsequent ecological analyses. Subsequent analyses could integrate multi-temporal UAV observations with hydrological conditions, prey-resource distribution, vessel activity,

underwater acoustic conditions, shoreline morphology, and other habitat covariates to quantify associations between porpoise detection patterns and environmental conditions and further investigate potential habitat-use patterns and spatial preferences [43,44].

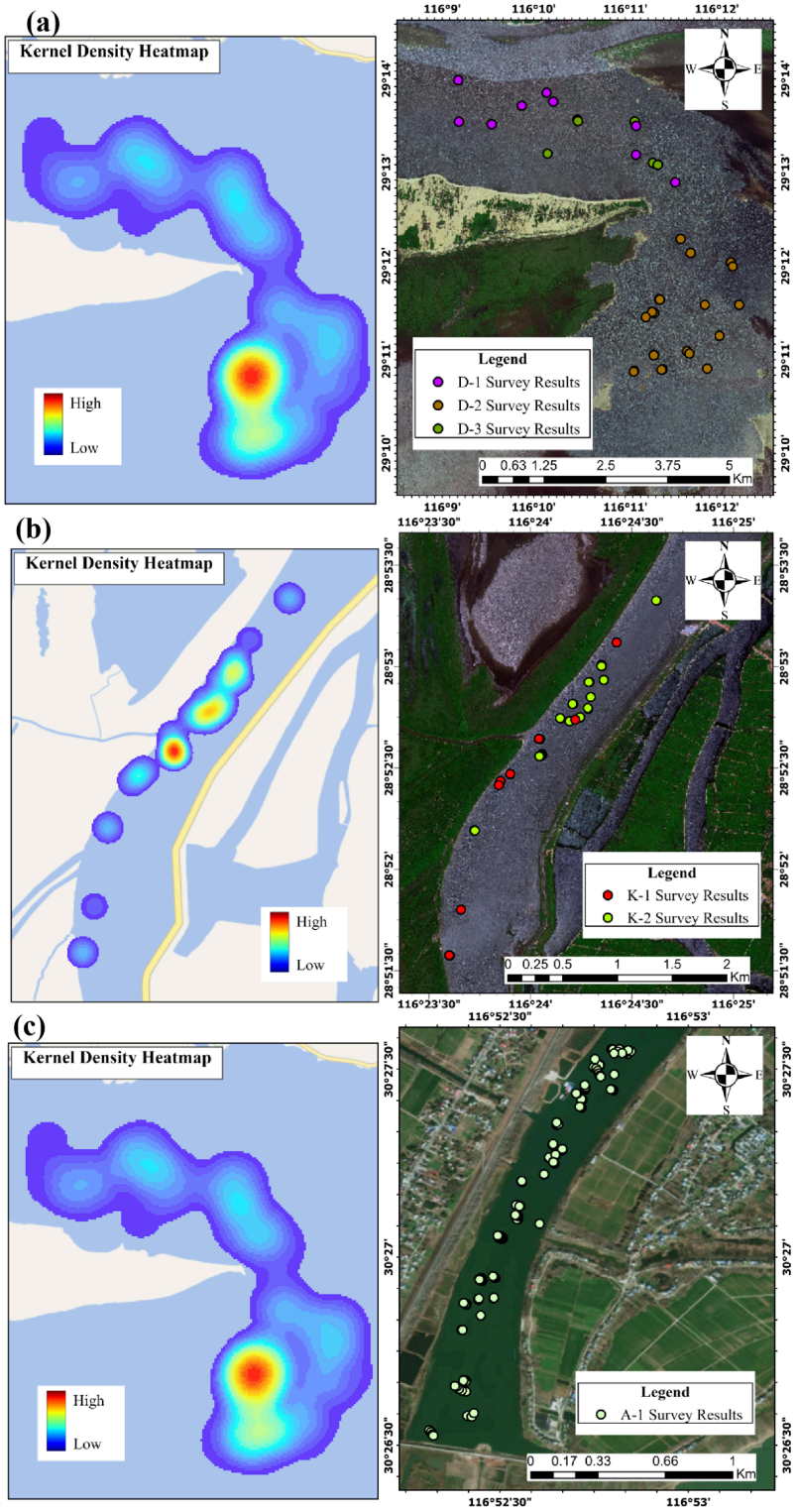


**Figure 12.** Spatial localization results of Yangtze finless porpoise and corresponding Kernel Density Estimation (KDE) maps in the three study subregions.

### *5.3. Effect of Imaging Time and Implications for Survey Design*

UAV data acquisition spanned from early morning to late afternoon and covered multiple periods of daytime porpoise activity. As shown in Figure 13a, missions in the

Duchang and Kangshan subregions were conducted between 05:47 and 11:42, whereas the Anqing mission was conducted between 16:18 and 17:32. Porpoise detections were obtained throughout the survey window, indicating that UAV imagery can capture surfacing events across a broad diurnal period. Mission-level variation in the proportion of images containing porpoises further indicated temporal differences in the frequency of recorded surfacing events. However, this indicator reflects both actual porpoise occurrence and image detectability and therefore cannot be interpreted as a direct measure of model performance.

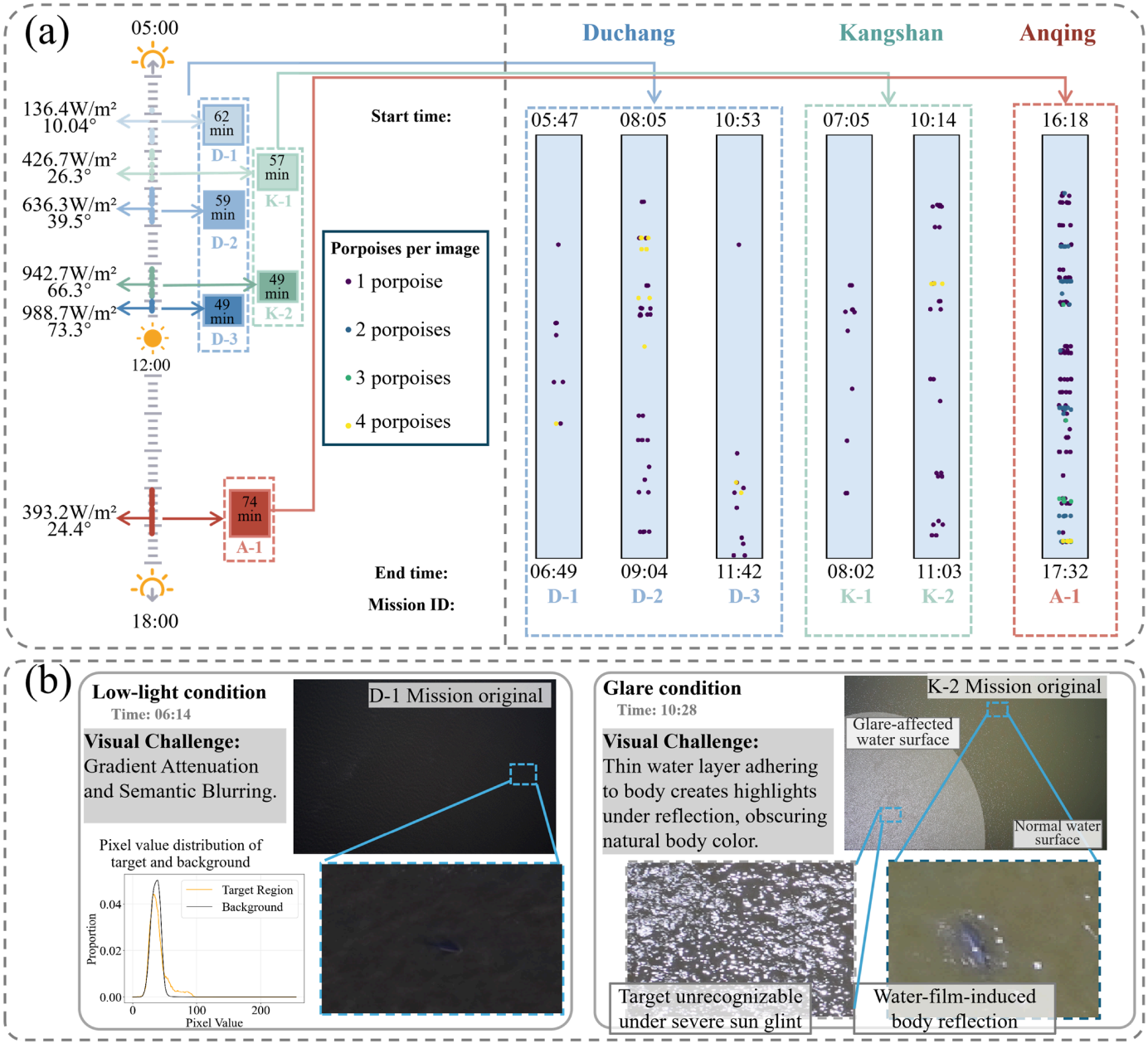


**Figure 13.** Influence of temporal factors on porpoise imaging and detection. (**a**) Time distribution of all survey missions and the number of porpoises detected per image in different subregions. (**b**) Representative examples of low-light and glare conditions, showing their effects on target visibility and background interference.

To quantitatively examine the influence of imaging time, the five missions conducted within the Poyang Lake system were arranged chronologically as D-1, K-1, D-2, K-2, and D-3. Their corresponding porpoise-image ratios were 0.28%, 0.47%, 0.80%, 0.79%,

and 0.25%, respectively, showing an overall increase from early morning to the middle-morning period, followed by a marked decline near noon. Clear-sky solar irradiance and solar elevation angle at the midpoint of each mission were estimated using the Pysolar library based on the acquisition time and geographic coordinates. Across the five missions, estimated solar irradiance increased from 136.43 to 426.73, 636.26, 942.65, and 988.70 W $m^{-2}$, while the solar elevation angle increased from 10.04° to 26.35°, 39.48°, 66.30°, and 73.27°, respectively. The relatively low porpoise-image ratio during D-1 may be associated with insufficient early-morning illumination, whereas the sharp decrease during D-3 may be related to stronger water-surface reflection and sun glint under high irradiance and a high solar elevation angle. These patterns suggest that intermediate morning periods may provide more favorable imaging conditions than either very early morning or near-noon periods.

The representative examples in Figure 13b further illustrate these effects. Under low-light conditions, the water surface appears darkened and the pixel-value distribution of the target region overlaps strongly with that of the background, indicating poor separability between porpoise and water-surface signals. The porpoise body therefore becomes visually inconspicuous and target visibility is substantially reduced. Under glare conditions, the image can be clearly divided into normal and glare-affected water-surface regions. Within areas of severe sun glint, the porpoise target becomes almost unrecognizable, whereas targets located outside or near the margin of the reflective region remain detectable. The enlarged example also suggests that water-film reflection on the porpoise body may further intensify local highlights and obscure natural body texture. Appropriate flight timing is therefore essential for balancing target visibility against background interference. Relative to the extreme conditions of early low light and strong sun glint, periods with more uniform illumination are more favorable for stable porpoise detection and subsequent spatial localization.

### *5.4. Limitations and Future Work*

Several limitations of the present study should be acknowledged. The present study was conducted using 648 image tiles, reflecting both the rarity of the Yangtze finless porpoise and the brief and infrequent nature of its surfacing events. To strengthen model reliability under this sampling constraint, data augmentation and repeated group-constrained dataset partitioning were adopted. For YOLO26-Lap, data augmentation increased Precision from $0.678 \pm 0.082$ to $0.847 \pm 0.003$, Recall from $0.688 \pm 0.127$ to $0.785 \pm 0.014$, F1-score from $0.682 \pm 0.102$ to $0.815 \pm 0.009$, and mAP@0.5 from $0.691 \pm 0.121$ to $0.880 \pm 0.150$. In addition, three group-constrained dataset partitions were generated. For each partition, two training runs with different random seeds were conducted, and their performance metrics were first averaged to obtain a partition-level result. The resulting three partition-level values were then summarized as the mean and standard deviation across the three partitions. This design reduced dependence on any single dataset split while accounting for stochastic variation during model training, thereby providing a more stable assessment of model performance. Future surveys covering additional seasons and observation periods will further expand the diversity of porpoise appearances and water-surface conditions.

The flight-segment -based hybrid georeferencing strategy also provides practical applicability in heterogeneous river and nearshore environments. In river sections such as Anqing and Kangshan, flight segments containing sufficient land overlap and stable feature matches can be refined using AAT-BA, substantially improving exterior-orientation estimation and spatial localization. For open-water segments with insufficient stable features, DG maintains continuous positioning capability. This combination allows the workflow to accommodate both feature-rich nearshore river sections and low-texture open-

water environments within the same UAV mission. Future work will further examine the transferability of the 10% land-overlap and 80-inlier criteria across different flight heights, overlap settings, waterbody types, and seasonal conditions.

Because independently surveyed ground-truth coordinates cannot be obtained for freely moving porpoises at the exact moment of surfacing, localization accuracy was evaluated through repeated observations of static verification points. The post-AAT-BA RMSE values of 0.62 m under straight-flight conditions and 0.67 m under turning-flight conditions quantify the geometric accuracy of the imaging and georeferencing workflow and provide a practical reference for the expected spatial localization uncertainty of detected porpoises. Future validation using controlled moving surface targets, synchronous water-level and wave observations, and additional flight geometries will support a more comprehensive assessment of target-level localization accuracy.

## 6. Conclusions

An integrated UAV-based workflow was implemented, combining oriented-bounding-box detection with LapConv-enhanced edge sensitivity and flight-segment-based hybrid DG/AAT-BA georeferencing, for the monitoring of surfacing Yangtze finless porpoises. The framework introduces three main innovations: (1) a lightweight, plug-and-play LapConv module that amplifies weak edge and low-contrast features of partially exposed small aquatic targets, improving detection robustness across heterogeneous water environments; (2) a flight-line-based hybrid georeferencing strategy that applies DG or AAT-BA according to predefined land-overlap and feature-matching criteria, enabling sub-meter static-verification-point localization in flight segments suitable for AAT-BA while retaining DG-based positioning capability over low-texture open water; and (3) an end-to-end integration of detection and localization, allowing rapid transformation of UAV imagery into actionable ecological data.

Evaluation across three group-constrained partitions showed that LapConv consistently improved the mean mAP@0.5 across all four YOLO-OBB models, with gains ranging from 0.019 to 0.026. YOLO11-Lap achieved the highest mean mAP@0.5 of 0.927, while YOLO26 showed the largest improvement in Precision. For static-verification-point localization, AAT-BA reduced RMSE from 2.59 to 0.62 m at VP1 and from 3.61 to 0.67 m at VP2 relative to DG. Kernel density analysis visualized distinct spatial distribution characteristics among the three survey subregions, including relatively dispersed detections in Duchang, a left-bank concentration in Kangshan, and an elongated concentration belt in Anqing.

This workflow is reproducible, minimally invasive, and readily transferable to other small freshwater cetaceans or aquatic wildlife monitoring applications. Beyond conventional UAV imagery, future extensions could include integration with passive acoustic monitoring, multi-season and multi-habitat sampling, and real-time onboard detection to enable adaptive survey strategies. Collectively, these innovations advance the UAV-based wildlife monitoring toolbox, providing a practical platform for georeferenced wildlife surveys and future fine-scale habitat analysis.

**Author Contributions:** Conceptualization and methodology by D.Y. and F.W.; data acquisition by Z.W. and T.C.; writing—original draft preparation by D.Y. and F.W.; writing—review and editing by F.W., W.R., Y.L. and S.Z.; project administration and funding acquisition by F.W. and S.Z.; All authors have read and agreed to the published version of the manuscript.

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**Data Availability Statement:** The data that support the findings of this study are available from the corresponding author upon reasonable request.

**Conflicts of Interest:** Author Dongxu Yang is a graduate student at the East China Sea Fisheries Research Institute, Chinese Academy of Fishery Sciences. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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